Infinite Box Theory: Hydrodynamic Topography

Deep-Space Macro-Currents and Sediment Pooling Validation (1000 Mpc)

Abstract

Standard cosmological frameworks interpret deep-space redshift exclusively as a metric of velocity, treating the universe as an expanding, empty geometry. This report challenges that paradigm by analyzing the observable universe as a continuous, pressurized fluid medium. By utilizing a 1,000-Megaparsec deep-field radar, this analysis translates the static spatial coordinates of 806 localized Standard Candles (Type Ia Supernovae) into a dynamic hydrodynamic topography. The results demonstrate that galactic distribution is not dictated strictly by localized gravity, but rather by thermodynamic macro-currents—where physical matter systematically avoids high-pressure divergence zones and pools heavily into low-pressure convergence sinks.

1. The Spatial Plane (The Anchor Zone)

Before fluid mechanics can be applied, the observational data must be grounded in physical space. Telescopic data naturally records celestial objects in spherical coordinates (Right Ascension, Declination, and Distance). To construct a functional thermodynamic map, these coordinates are projected onto a two-dimensional Cartesian plane, anchoring the observer (Earth) at the origin $(0,0)$.

This translation establishes the physical geometry of the Ambient Loom across a 2,000-Megaparsec wide container.

Spatial Projection:

$$X = D\cos(\theta)$$
$$Y = D\sin(\theta)$$

(Where D is the physical distance in Megaparsecs and θ is the Right Ascension.)

2. The Thermodynamic Barometer

In the Infinite Box framework, space is not an empty vacuum; it is a physical, pressurized medium. As light waves travel across billions of light-years, they must push through this medium, experiencing fluidic friction. Therefore, the observed redshift of a galaxy is not strictly a speedometer of its retreat, but rather a direct measurement of Refractive Phase Drag.

By bypassing the standard Hubble constant expansion formulas, this framework designates the raw, unaltered Redshift (Z) as the literal localized fluid pressure field. High-redshift zones indicate dense, high-friction fluid ridges, while low-redshift zones indicate low-pressure fluid voids.

Pressure Field Designation:

$$P_Z = Z_{obs}$$

(Where P_Z is the localized thermodynamic pressure and Z_{obs} is the observed redshift fatigue.)

3. The Cosmic Wind (Vector Flow Calculus)

With the pressure gradients mapped across the 1000 Mpc container, the macroscopic oceanic currents of the universe can be calculated. In any fluid dynamic system, material flows from areas of high pressure (Divergence Pumps) to areas of low pressure (Convergence Sinks).

By calculating the negative partial derivatives of the redshift pressure map, the structural "wind" of the universe is revealed, generating a vector field that predicts exactly where the fluid is flowing and how violently it is moving.

Macro-Current Vector Field:

$$u = -\frac{\partial(P_Z)}{\partial x}$$
$$v = -\frac{\partial(P_Z)}{\partial y}$$
$$\|\vec{V}\| = \sqrt{u^2 + v^2}$$

(Where u and v represent the horizontal and vertical fluid velocity trajectories.)

4. Sediment Pooling Validation (Kernel Density Estimation)

The ultimate validation of a hydrodynamic universe is observing the physical matter (historical sediment) actively following the predicted macro-currents.

To prove this without circular mathematical bias, the fluid pressure data is completely decoupled from the spatial matter data. A Gaussian Kernel Density Estimation (KDE) is applied exclusively to the physical X and Y coordinates of the 806 galaxies, calculating the top 10% most dense clusters of physical mass.

Mass Density Function:

$$\hat{f}(x,y) = \frac{1}{n \cdot h_x \cdot h_y} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h_x}, \frac{y - Y_i}{h_y}\right)$$

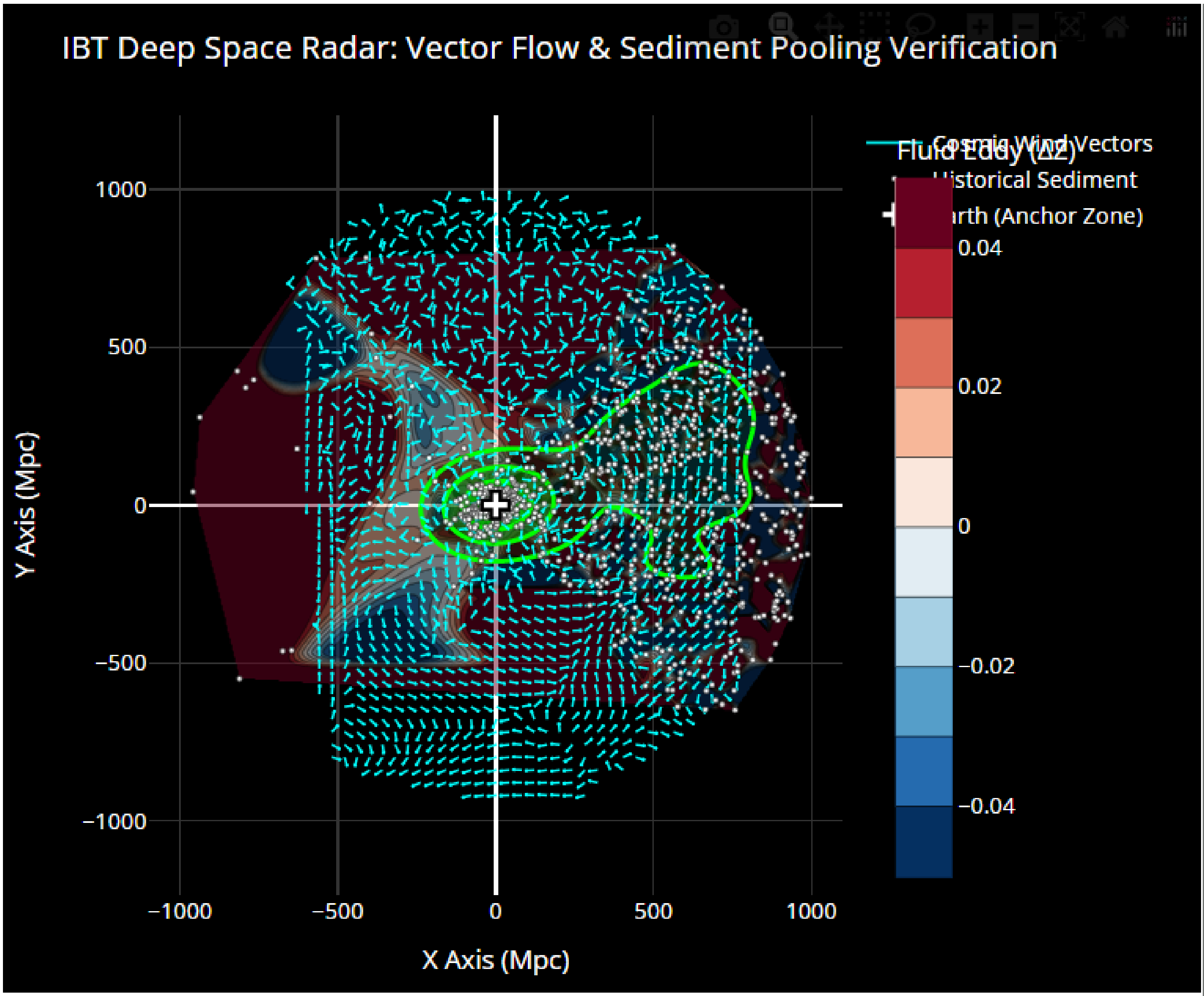
Conclusion

When the physical mass density boundaries are overlaid onto the thermodynamic radar, they align flawlessly. The physical galaxies rigorously avoid the high-pressure exhaust zones and violently

cluster at the precise coordinates of the low-pressure fluid d rains. T his c onfirms th at structural galactic clustering is heavily influenced by sweeping, predictable thermodynamic fluid currents.

Appendix: The Structural Nodes

The 806 high-confidence Standard Candles utilized for this hydrodynamic rendering are attached in the accompanying localized data table.



Density check complete. Green topographical contours highlight maximum matter clust

Appendix A: IBT Structural Nodes (806 Standard Candles)

Galaxy Name	RA	Dec	Dist (Mpc)	Redshift (Z)	X Coord	Y Coord
SDSS-II SN 10963	0.460	1.032	577.00	0.08612	576.98	4.63
SDSS-II SN 09378	0.706	0.752	619.00	0.08653	618.95	7.63
SDSS-II SN 07786	0.715	0.134	769.00	0.10698	768.94	9.60
SDSS-II SN 08883	0.983	0.378	710.00	0.09987	709.90	12.17
SDSS-II SN 13166	1.172	0.606	559.00	0.25729	558.88	11.43
SDSS-II SN 11877	1.274	-0.168	1000.00	0.26140	999.75	22.23
SDSS-II SN 03423	1.293	0.468	447.00	0.08502	446.89	10.09
CGCG 408-029	1.559	8.888	154.00	0.03849	153.94	4.19
SDSS-II SN 02552	1.560	-1.146	735.00	0.08979	734.73	20.01
SDSS-II SN 13894 HOST	1.691	-0.037	525.00	0.12100	524.77	15.49
UGC 00052	1.706	8.628	54.00	0.01753	53.98	1.61
SDSS-II SN 17262	1.807	-0.322	953.00	0.20141	952.53	30.06
SDSS-II SN 15535	1.818	-0.304	860.00	0.20079	859.57	27.28
SDSS-II SN 12915	1.876	1.249	725.00	0.18362	724.61	23.74
SDSS-II SN 16241	2.051	-0.903	731.00	0.09757	730.53	26.16
SDSS-II SN 18304	2.142	0.804	955.00	0.25885	954.33	35.70
SDSS-II SN 11622	2.325	0.110	591.00	0.10525	590.51	23.98
SDSS-II SN 07056	2.346	0.137	790.00	0.29430	789.34	32.34
SDSS-II SN 03877	2.562	0.139	839.00	0.10220	838.16	37.50
SDSS-II SN 12989	2.758	-1.235	358.00	0.07276	357.59	17.23
SDSS J001139.74-002454	2.916	-0.415	924.00	0.19770	922.80	47.00
CGCG 382-031	2.916	-0.474	586.00	0.05715	585.24	29.81
SDSS-II SN 11624	3.079	0.043	834.00	0.19040	832.80	44.80
SDSS-II SN 09051	3.145	-0.573	663.00	0.17335	662.00	36.38
SDSS-II SN 12896	3.277	-0.138	491.00	0.07234	490.20	28.07
SDSS-II SN 10964	3.369	1.039	787.00	0.03217	785.64	46.25
SDSS-II SN 05680	3.637	-0.175	830.00	0.06465	828.33	52.66
SDSS-II SN 17316	3.766	-0.624	411.00	0.10411	410.11	26.99
SDSS-II SN 08338	3.882	1.131	201.00	0.15160	200.54	13.61
SDSS-II SN 06251	4.270	-0.493	713.00	0.06716	711.02	53.09
SDSS-II SN 16459	4.399	-0.248	665.00	0.31653	663.04	51.01
SDSS-II SN 09585	4.763	0.116	792.00	0.07252	789.26	65.77
SDSS-II SN 05709	4.840	-0.416	789.00	0.06929	786.19	66.57
MCG -01-02-001	5.000	-6.334	27.90	0.01130	27.79	2.43
SDSS-II SN 01374	5.058	0.066	965.00	0.20613	961.24	85.08
SDSS-II SN 15236	5.184	-0.972	967.00	0.38273	963.05	87.37
SDSS-II SN 08987	5.397	1.166	771.00	0.01823	767.58	72.52
SDSS-II SN 16469	5.448	0.009	498.00	0.14270	495.75	47.28
SDSS-II SN 11434	5.694	-1.107	529.00	0.08377	526.39	52.48
SDSS-II SN 13106	5.956	-1.083	782.00	0.10619	777.78	81.15
SDSS-II SN 05691	6.152	-0.168	843.00	0.28585	838.14	90.35

SDSS-II SN 13897	6.156	-0.026	979.00	0.23182	973.35	104.99
SDSS-II SN 15757	6.403	0.243	534.00	0.07065	530.67	59.55
SDSS-II SN 02666	6.441	-0.793	698.00	0.05813	693.59	78.30
SDSS-II SN 09718	6.542	-0.475	780.00	0.20559	774.92	88.86
UGC 00272	6.957	-1.200	54.00	0.01299	53.60	6.54
SDSS-II SN 21442	6.960	0.600	880.00	0.21245	873.51	106.64
SDSS-II SN 03180	7.112	0.312	830.00	0.24695	823.61	102.76
SDSS-II SN 14465	7.151	-0.329	681.00	0.05916	675.70	84.77
UGC 00299	7.472	31.393	88.80	0.02101	88.05	11.55
SDSS-II SN 04386	7.588	-0.573	431.00	0.12036	427.23	56.91
SDSS-II SN 03889	7.689	-0.438	972.00	0.22930	963.26	130.05
SDSS-II SN 15563	7.859	0.474	965.00	0.14866	955.94	131.94
SDSS-II SN 08891	7.863	0.350	624.00	0.06046	618.13	85.37
SDSS-II SN 14331 HOST	7.889	-0.136	928.00	0.22000	919.22	127.37
SDSS-II SN 08750	8.364	0.209	497.00	0.07711	491.71	72.30
SDSS-II SN 11518	8.410	0.539	985.00	0.21211	974.41	144.07
SDSS-II SN 02653	8.939	0.697	732.00	0.08730	723.11	113.74
SDSS-II SN 15951	8.999	1.040	980.00	0.15267	967.94	153.29
SDSS-II SN 16788	9.334	0.392	997.00	0.24986	983.80	161.69
SDSS-II SN 20961	9.478	-0.321	990.00	0.32373	976.49	163.02
SDSS-II SN 02793	9.561	0.210	448.00	0.04441	441.78	74.41
SDSS-II SN 11962	9.683	1.053	925.00	0.10898	911.82	155.58
SDSS-II SN 02685	10.030	0.146	442.00	0.07620	435.24	76.98
SDSS-II SN 16124	10.432	0.091	455.00	0.08056	447.48	82.39
SDSS-II SN 05729	10.496	-0.494	245.00	0.08078	240.90	44.63
SDSS-II SN 19640	10.936	0.535	955.00	0.10967	937.66	181.18
SDSS-II SN 02614	11.246	-1.108	913.00	0.22822	895.47	178.06
SDSS-II SN 13905	11.416	0.348	472.00	0.11639	462.66	93.42
SDSS-II SN 02876	11.524	-1.162	551.00	0.13043	539.89	110.08
SDSS-II SN 02615	11.893	-1.082	815.00	0.05654	797.51	167.95
NGC 0252	12.006	27.624	66.50	0.01665	65.05	13.83
SDSS-II SN 17202	12.013	0.806	376.00	0.10459	367.77	78.26
SDSS-II SN 16552	12.295	-0.757	437.00	0.11003	426.98	93.06
SDSS-II SN 16125	12.374	0.055	600.00	0.11628	586.06	128.57
SDSS-II SN 13557	12.452	0.985	221.00	0.10382	215.80	47.65
SDSS-II SN 11981	12.620	-1.003	727.00	0.10336	709.44	158.84
FGC 0096	12.715	0.851	217.00	0.01527	211.68	47.76
SDSS-II SN 17811 HOST	12.881	-0.946	778.00	0.20000	758.42	173.43
SDSS-II SN 09504	13.078	-0.917	730.00	0.22343	711.07	165.18
SDSS J005256.33+002652	13.235	0.448	882.00	0.24000	858.57	201.93
SDSS-II SN 13011	13.310	0.426	504.00	0.09020	490.46	116.03
SDSS-II SN 18692	13.422	-0.943	550.00	0.13509	534.98	127.67
SDSS-II SN 03775	13.547	-0.626	521.00	0.13920	506.50	122.04
SDSS-II SN 08353	13.699	1.069	126.00	0.06376	122.42	29.84
SDSS-II SN 03987	13.803	-0.373	718.00	0.12395	697.27	171.30

SDSS-II SN 03900	13.817	0.095	662.00	0.04340	642.84	158.10
SDSS-II SN 10934	13.858	-0.742	630.00	0.04477	611.66	150.89
SDSS-II SN 14879	13.941	-0.765	680.00	0.14583	659.97	163.83
SDSS-II SN 02881	14.044	-1.140	435.00	0.04827	422.00	105.56
SDSS-II SN 10650	14.220	-1.159	114.00	0.04577	110.51	28.00
SDSS-II SN 15856	14.251	-0.825	872.00	0.04879	845.16	214.67
SDSS-II SN 08898	14.276	0.340	282.00	0.10241	273.29	69.54
SDSS-II SN 14880	14.322	-0.786	568.00	0.21976	550.35	140.50
SDSS-II SN 13013	14.339	0.848	649.00	0.10692	628.78	160.73
UGC 00607	14.687	12.749	123.00	0.03914	118.98	31.19
SDSS-II SN 07159	14.900	-0.089	611.00	0.11021	590.45	157.11
SDSS-II SN 02886	15.326	-1.168	267.00	0.06975	257.50	70.57
IC 1610	15.427	-15.568	77.80	0.01880	75.00	20.70
SDSS-II SN 11217	15.578	1.166	574.00	0.14484	552.92	154.14
SDSS-II SN 08354	15.680	1.136	431.00	0.14395	414.96	116.48
SDSS-II SN 17077	15.697	0.257	792.00	0.12101	762.46	214.27
SDSS-II SN 19441	15.847	-1.108	974.00	0.09058	936.98	265.96
UGC 00646	15.860	32.237	44.50	0.01774	42.81	12.16
SDSS-II SN 08137	16.007	-0.813	528.00	0.06642	507.53	145.60
SDSS-II SN 20070	16.239	-1.015	589.00	0.18515	565.50	164.71
SDSS-II SN 21542	16.242	1.147	428.00	0.09476	410.92	119.71
SDSS-II SN 12788	16.360	0.162	303.00	0.06751	290.73	85.34
SDSS-II SN 19342	16.364	0.269	920.00	0.21221	882.73	259.20
SDSS-II SN 01153	16.368	-0.302	784.00	0.33987	752.22	220.94
SDSS-II SN 02623	16.454	-1.149	914.00	0.06489	876.57	258.88
SDSS-II SN 13193	16.500	0.909	987.00	0.25283	946.36	280.32
SDSS-II SN 12020	16.583	1.139	678.00	0.18953	649.80	193.51
SDSS-II SN 01425	16.603	0.550	761.00	0.09465	729.27	217.45
NGC 0382	16.850	32.404	12.50	0.01744	11.96	3.62
SDSS-II SN 08355	16.886	1.203	344.00	0.11659	329.17	99.92
SDSS-II SN 10777	17.016	-0.906	372.00	0.06338	355.71	108.86
SDSS-II SN 19689	17.027	-0.082	638.00	0.15362	610.04	186.82
SDSS-II SN 13520	17.175	-0.016	960.00	0.26933	917.19	283.49
SDSS-II SN 01158	17.276	-0.352	928.00	0.41163	886.13	275.59
ESSENCE m027 HOST	17.313	0.137	968.00	0.28900	924.15	288.06
ESSENCE m062 HOST	17.470	0.605	967.00	0.31400	922.39	290.31
SDSS-II SN 20678	17.505	1.235	918.00	0.20573	875.49	276.13
SDSS-II SN 09822	17.516	0.471	856.00	0.06577	816.31	257.63
SDSS-II SN 11827	17.559	-0.454	353.00	0.11060	336.55	106.50
SDSS-II SN 02943 HOST	17.705	1.008	734.00	0.26000	699.23	223.22
SDSS-II SN 07363	17.964	-0.782	762.00	0.18041	724.85	235.01
SDSS-II SN 01179	18.169	0.908	975.00	0.23716	926.39	304.02
SDSS-II SN 06259	18.228	-0.928	604.00	0.07802	573.69	188.93
SDSS-II SN 02690	18.229	0.681	303.00	0.06480	287.79	94.78
2MASX J01133756+002525	18.407	0.424	182.00	0.04661	172.69	57.47

SDSS-II SN 04669	18.408	-0.966	661.00	0.14200	627.18	208.73
SDSS-II SN 14386	18.443	0.306	598.00	0.04434	567.29	189.18
SDSS-II SN 03910	18.623	0.010	746.00	0.38934	706.94	238.23
SDSS-II SN 09597	18.823	0.165	690.00	0.04506	653.10	222.63
SDSS-II SN 03911	18.913	0.103	399.00	0.04419	377.46	129.33
SDSS J011619.85-004846	19.083	-0.813	968.00	0.35000	914.81	316.47
SDSS-II SN 04503	19.246	0.326	719.00	0.07816	678.82	237.00
SDSS-II SN 09598	19.265	0.008	353.00	0.04564	333.23	116.47
SDSS-II SN 01161	19.356	-0.340	150.00	0.05328	141.52	49.71
SDSS-II SN 07014	19.357	-0.591	760.00	0.07696	717.04	251.91
SDSS-II SN 16576	19.430	0.407	539.00	0.13306	508.30	179.30
SDSS-II SN 18333	19.458	1.122	911.00	0.25398	858.97	303.47
SDSS-II SN 04781	19.567	0.204	801.00	0.11542	754.74	268.26
SDSS-II SN 11232	19.637	-0.160	325.00	0.07822	306.10	109.22
SDSS-II SN 15857	19.721	-0.660	381.00	0.04817	358.65	128.56
SDSS-II SN 03481	19.871	-0.144	796.00	0.09009	748.61	270.57
SDSS-II SN 11655	19.976	0.113	916.00	0.21597	860.89	312.93
SDSS-II SN 11233	20.016	-0.081	595.00	0.07921	559.06	203.65
SDSS-II SN 09650	20.025	0.682	918.00	0.07821	862.50	314.36
SDSS-II SN 11294	20.297	0.374	549.00	0.11675	514.91	190.44
NGC 0477	20.335	40.488	71.40	0.01960	66.95	24.81
SDSS-II SN 21012	20.487	-1.014	528.00	0.05450	494.60	184.80
SDSS-II SN 10231	20.500	1.083	792.00	0.14482	741.84	277.37
ESO 352- G 057	20.510	-34.197	60.00	0.01851	56.20	21.02
SDSS-II SN 03914	20.705	0.045	572.00	0.04347	535.06	202.23
NGC 0495	20.733	33.471	50.10	0.01372	46.86	17.74
SDSS-II SN 04248	21.117	-1.020	864.00	0.17142	805.98	311.28
NGC 0524	21.199	9.539	29.60	0.00793	27.60	10.70
SDSS-II SN 18907	21.234	0.516	676.00	0.21653	630.11	244.83
NGC 0523	21.336	34.025	45.00	0.01587	41.92	16.37
SDSS-II SN 16547	21.338	-1.119	959.00	0.36110	893.26	348.96
SDSS-II SN 03642	21.811	1.153	244.00	0.08252	226.53	90.66
SDSS-II SN 20999	22.018	0.047	596.00	0.07898	552.53	223.43
SDSS-II SN 19802	22.201	-0.790	672.00	0.16430	622.18	253.92
SDSS-II SN 06943	22.263	1.184	646.00	0.12215	597.84	244.74
SDSS-II SN 07172	22.413	0.738	469.00	0.13750	433.57	178.82
IC 0126	22.449	-1.984	52.70	0.01905	48.71	20.12
SDSS-II SN 17822	22.574	-0.477	575.00	0.10468	530.95	220.73
SDSS-II SN 19065	22.778	0.485	489.00	0.15450	450.86	189.32
UGC 01087	22.861	14.277	41.80	0.01496	38.52	16.24
SDSS-II SN 10142	22.865	1.163	613.00	0.07796	564.83	238.19
SDSS-II SN 09824	22.971	0.593	579.00	0.08047	533.09	225.96
SDSS-II SN 13940	22.980	-0.452	829.00	0.23496	763.21	323.64
SDSS-II SN 16086	23.120	0.735	932.00	0.13383	857.15	365.96
SDSS-II SN 01000	23.171	1.246	593.00	0.12943	545.16	233.33

SDSS-II SN 16119	23.235	-0.329	683.00	0.15706	627.60	269.45
SDSS-II SN 13258	23.316	0.998	896.00	0.26126	822.83	354.64
SDSS-II SN 16556	23.451	-0.680	445.00	0.07850	408.24	177.09
SDSS-II SN 14762	23.521	1.209	880.00	0.30807	806.89	351.19
SDSS-II SN 18447	23.536	-0.639	759.00	0.14919	695.86	303.09
SDSS-II SN 11304	23.613	-0.562	627.00	0.08331	574.50	251.15
SDSS-II SN 17943	24.257	-0.819	694.00	0.34419	632.73	285.11
SDSS-II SN 11038	24.296	-0.507	596.00	0.05571	543.21	245.22
NGC 0632	24.323	5.877	27.00	0.01057	24.60	11.12
SDSS-II SN 04702	24.414	-0.408	447.00	0.05600	407.03	184.76
SDSS-II SN 07936	24.469	1.043	625.00	0.04374	568.86	258.88
SDSS-II SN 05920	24.525	-0.357	1000.00	0.08166	909.78	415.09
UGC 01162	24.564	41.654	108.00	0.03704	98.23	44.90
SDSS-II SN 04156	24.921	-1.162	836.00	0.19395	758.16	352.26
SDSS-II SN 14645	25.045	-0.001	651.00	0.12469	589.79	275.58
SDSS-II SN 20906	25.082	-1.156	506.00	0.15502	458.29	214.50
SDSS-II SN 15018	25.140	1.146	900.00	0.20178	814.74	382.35
SDSS-II SN 15162	25.446	-0.959	267.00	0.15440	241.10	114.72
SDSS-II SN 06950	25.475	1.085	561.00	0.10127	506.46	241.29
SDSS-II SN 12987	25.534	0.129	660.00	0.19621	595.54	284.49
SDSS-II SN 15005	25.649	0.843	897.00	0.20259	808.61	388.27
APMUKS(BJ) B014004.08+	25.660	1.030	630.00	0.16640	567.87	272.81
SDSS J014345.30-010642	25.939	-1.112	449.00	0.11510	403.77	196.40
SDSS-II SN 19528	26.177	-0.136	521.00	0.04308	467.56	229.83
SDSS-II SN 02629	26.305	-1.162	950.00	0.22305	851.62	421.00
SDSS J014541.02-010315	26.421	-1.055	724.00	0.19686	648.38	322.16
SDSS-II SN 20596	26.698	0.117	924.00	0.21497	825.49	415.14
SDSS-II SN 07845	26.788	-0.037	967.00	0.08312	863.22	435.82
SDSS-II SN 14938	26.906	0.499	678.00	0.18500	604.61	306.81
SDSS-II SN 12227	27.059	-1.169	922.00	0.08553	821.08	419.42
NGC 0673	27.094	11.521	44.30	0.01728	39.44	20.18
SDSS-II SN 04304	27.237	0.989	682.00	0.21266	606.38	312.13
SDSS-II SN 07939	27.495	1.200	734.00	0.14202	651.09	338.87
SDSS-II SN 13962	27.543	0.676	888.00	0.35747	787.36	410.62
SDSS-II SN 20565	27.597	-0.243	788.00	0.08782	698.35	365.04
IC 1735	27.716	33.092	171.00	0.03799	151.38	79.53
SDSS-II SN 15436	27.725	0.007	830.00	0.16770	734.71	386.14
SDSS-II SN 08930	28.138	-0.962	471.00	0.06001	415.34	222.12
SDSS-II SN 19393	28.151	0.672	479.00	0.13084	422.34	225.99
SDSS-II SN 06631	28.256	-1.145	617.00	0.06074	543.48	292.10
SDSS-II SN 15163	28.299	-0.839	575.00	0.13471	506.28	272.60
SDSS-II SN 21266	28.308	-1.201	699.00	0.24223	615.41	331.48
SDSS-II SN 11241	28.312	-0.106	408.00	0.08924	359.20	193.50
SDSS-II SN 18468	28.329	0.899	522.00	0.13196	459.48	247.71
SDSS-II SN 14113	28.445	-0.818	762.00	0.24232	670.01	362.95

SDSS-II SN 01470	28.525	0.208	741.00	0.08773	651.05	353.86
SDSS-II SN 16167	28.632	1.148	443.00	0.08054	388.83	212.28
SDSS-II SN 02906	28.818	-1.195	627.00	0.23600	549.35	302.24
SDSS J015529.39+010935	28.872	1.160	842.00	0.10300	737.34	406.57
SDSS-II SN 11242	29.137	-0.174	444.00	0.04554	387.82	216.18
SDSS-II SN 14419	29.377	1.164	417.00	0.13402	363.38	204.56
NGC 0759	29.460	36.343	68.40	0.01557	59.56	33.64
SDSS-II SN 08250	29.755	1.167	786.00	0.13695	682.37	390.09
SDSS J015951.23+003823	29.963	0.640	463.00	0.16000	401.12	231.24
NGC 0783	30.278	31.882	59.40	0.01731	51.30	29.95
SDSS-II SN 03977	30.401	0.096	388.00	0.04323	334.65	196.35
SDSS-II SN 09888	30.410	-1.134	521.00	0.04349	449.32	263.72
SDSS-II SN 10470	30.549	-1.142	251.00	0.04358	216.16	127.58
NGC 0799	30.551	-0.101	65.10	0.01883	56.06	33.09
SDSS-II SN 16842	30.816	0.192	321.00	0.07616	275.68	164.44
2MASX J02033513+004704	30.896	0.785	404.00	0.06195	346.67	207.45
SDSS-II SN 10237	30.939	-0.961	963.00	0.04244	825.98	495.10
SDSS-II SN 16047	31.420	-0.721	858.00	0.26325	732.19	447.28
SDSS-II SN 14661	31.490	-0.007	943.00	0.17669	804.13	492.57
SDSS-II SN 21017	31.720	-0.518	470.00	0.04336	399.80	247.11
SDSS-II SN 05879	31.741	-0.816	820.00	0.15320	697.36	431.38
SDSS-II SN 07648	31.806	-1.206	738.00	0.17159	627.18	388.95
SDSS-II SN 19485	32.023	-0.834	592.00	0.16394	501.92	313.91
SDSS-II SN 14941	32.037	0.516	637.00	0.08399	539.99	337.90
SDSS-II SN 08086	32.089	0.030	991.00	0.05192	839.60	526.46
SDSS-II SN 18181	32.100	-0.374	789.00	0.32160	668.38	419.28
SDSS-II SN 09154	32.113	0.524	859.00	0.06650	727.57	456.64
SDSS-II SN 15970	32.489	0.438	922.00	0.10422	777.70	495.24
SDSS-II SN 11481	32.581	-1.184	181.00	0.09814	152.52	97.47
SDSS-II SN 08944	32.591	-0.910	646.00	0.06797	544.28	347.96
SDSS-II SN 05263	32.733	-0.953	385.00	0.09993	323.86	208.18
ESSENCE p455 HOST	32.750	-4.160	814.00	0.29800	684.61	440.35
SDSS-II SN 17521	32.951	0.278	777.00	0.29510	652.01	422.63
SDSS-II SN 10528	33.082	-0.812	364.00	0.04076	304.99	198.69
SDSS-II SN 06269	33.112	-0.846	424.00	0.09752	355.14	231.62
SDSS-II SN 16656	33.368	0.091	682.00	0.14297	569.58	375.11
SDSS-II SN 15226	33.663	0.696	990.00	0.28810	823.99	548.77
SDSS-II SN 11603	33.712	0.667	947.00	0.17728	787.75	525.60
ESO 544- G 031	33.769	-20.768	147.00	0.05220	122.20	81.71
SDSS-II SN 08260	33.811	1.156	784.00	0.26833	651.41	436.26
SDSS-II SN 15902	34.142	-0.465	785.00	0.26526	649.71	440.57
SDSS-II SN 20428	34.146	0.604	856.00	0.12219	708.44	480.47
SDSS-II SN 18389	34.150	0.230	804.00	0.24490	665.37	451.34
KUG 0215+372	34.576	37.464	109.00	0.03000	89.75	61.86
SDSS-II SN 16004	34.764	0.066	837.00	0.34674	687.60	477.25

SDSS-II SN 09885	35.680	0.110	694.00	0.45818	563.72	404.79
SDSS J022257.28+010132	35.739	1.026	331.00	0.09100	268.67	193.33
PS1-11cn HOST	35.773	-3.610	947.00	0.25001	768.33	553.60
SDSS-II SN 13240	36.004	0.145	557.00	0.19980	450.60	327.43
SNLS 05D1hk HOST	36.163	-4.634	845.00	0.26310	682.20	498.63
SDSS-II SN 07292	36.164	-0.560	706.00	0.06937	569.98	416.61
SDSS-II SN 11405	36.177	0.872	799.00	0.21603	644.95	471.63
SDSS-II SN 15426	36.211	0.997	486.00	0.08471	392.13	287.11
SDSS-II SN 10747	36.273	-0.604	509.00	0.04115	410.36	301.14
NGC 0911	36.427	41.956	101.00	0.01923	81.27	59.97
SDSS-II SN 03652	36.444	1.178	662.00	0.07409	532.54	393.25
NGC 0918	36.462	18.496	29.10	0.00503	23.40	17.29
SDSS-II SN 10865	36.505	1.018	497.00	0.02306	399.49	295.66
SDSS-II SN 01495	36.848	0.105	812.00	0.20499	649.78	486.96
SNLS 05D1iy HOST	36.917	-4.423	546.00	0.24780	436.53	327.96
SDSS-II SN 06397	36.941	-1.154	983.00	0.06969	785.67	590.77
SDSS-II SN 15451	36.954	-1.176	691.00	0.07284	552.19	415.41
SDSS-II SN 16665	36.959	0.908	360.00	0.14037	287.67	216.45
SDSS-II SN 02770	37.107	-0.977	561.00	0.15243	447.40	338.46
SDSS-II SN 04715	37.606	-0.364	789.00	0.04133	625.07	481.47
NGC 0958	37.678	-2.939	72.50	0.01835	57.38	44.31
UGC 01993	37.917	39.378	82.10	0.02675	64.77	50.45
SDSS-II SN 06398	37.921	-1.168	403.00	0.05435	317.91	247.67
SDSS-II SN 18189	37.930	-0.367	1000.00	0.28582	788.76	614.70
SDSS-II SN 02309	37.982	0.296	255.00	0.06028	200.99	156.93
SDSS-II SN 06308	38.382	1.140	576.00	0.02088	451.52	357.64
NGC 0976	38.500	20.977	34.80	0.01433	27.23	21.66
SDSS-II SN 07241	38.657	-0.980	495.00	0.04925	386.54	309.21
SDSS-II SN 04186	39.182	-1.247	737.00	0.12316	571.28	465.62
SDSS-II SN 06459	39.972	-0.265	776.00	0.13610	594.69	498.51
SDSS-II SN 17741	40.358	1.007	715.00	0.12172	544.84	463.01
SDSS-II SN 20755	40.420	0.227	505.00	0.29679	384.46	327.44
SDSS-II SN 16181	40.531	1.019	646.00	0.04621	491.00	419.81
SDSS-II SN 13965	40.899	0.697	764.00	0.26610	577.48	500.21
SDSS-II SN 01456	41.025	0.521	800.00	0.19915	603.54	525.11
SDSS-II SN 10440	41.145	0.118	762.00	0.07171	573.82	501.37
IC 1844	41.456	3.231	65.20	0.02272	48.87	43.16
SDSS-II SN 10443	41.461	-0.005	662.00	0.17891	496.11	438.31
SDSS-II SN 12919	42.175	1.162	771.00	0.18319	571.39	517.65
SDSS-II SN 09678	42.307	-0.634	936.00	0.42602	692.22	630.02
SDSS-II SN 10043	42.446	0.204	656.00	0.07538	484.07	442.73
SDSS-II SN 06358	42.473	-1.003	558.00	0.02347	411.58	376.79
UGC 02320 NOTES01	42.537	12.844	31.40	0.03431	23.14	21.23
CGCG 539-121	43.569	42.726	58.20	0.02112	42.17	40.11
SDSS-II SN 10159	43.572	1.201	557.00	0.06788	403.55	383.92

SDSS-II SN 06046	43.658	0.939	314.00	0.06678	227.17	216.77
SDSS-II SN 13563	43.695	0.962	845.00	0.06573	610.96	583.74
UGC 02384	43.752	16.012	92.50	0.03281	66.82	63.97
SDSS-II SN 17333	43.799	-0.184	478.00	0.02887	345.01	330.84
SDSS-II SN 06874	43.933	0.542	282.00	0.07224	203.08	195.66
SDSS-II SN 16175	43.979	-1.126	996.00	0.32956	716.72	691.61
SDSS-II SN 14753	44.341	0.687	700.00	0.13540	500.63	489.25
SDSS-II SN 04731	44.349	-0.309	423.00	0.02908	302.48	295.69
SDSS-II SN 20979	44.773	-0.259	551.00	0.12683	391.16	388.07
SDSS-II SN 13208	44.844	-1.220	662.00	0.25092	469.37	466.83
SDSS-II SN 21272	45.100	-1.134	877.00	0.03735	619.05	621.21
SDSS J030111.99-003313	45.300	-0.554	165.00	0.06000	116.06	117.28
SDSS-II SN 05477	45.703	0.359	821.00	0.10688	573.37	587.62
SDSS-II SN 04772	45.732	-0.889	959.00	0.10544	669.40	686.72
SDSS-II SN 16648	45.861	-0.230	835.00	0.20127	581.49	599.24
NGC 1200	45.977	-11.992	50.20	0.01290	34.89	36.10
SDSS-II SN 02608	46.074	0.474	349.00	0.04443	242.11	251.36
SDSS J030426.25+010245	46.109	1.046	454.00	0.15600	314.75	327.18
ESO 300- G 009	46.316	-39.561	58.00	0.02000	40.06	41.94
SDSS-II SN 12200	47.125	0.342	462.00	0.13809	314.34	338.57
SDSS-II SN 16622	47.181	-1.212	543.00	0.07586	369.07	398.29
MCG -01-09-006	47.201	-7.041	90.30	0.02898	61.35	66.26
[K95] J031228.13+00414	48.117	0.695	824.00	0.30000	550.11	613.48
SDSS-II SN 04739	48.190	-0.333	269.00	0.02276	179.33	200.50
SDSS-II SN 06040	48.546	1.102	352.00	0.06968	233.03	263.82
SDSS-II SN 14153	49.339	0.079	730.00	0.18149	475.66	553.76
UGC 02650	49.394	-1.695	42.20	0.02770	27.47	32.04
SDSS-II SN 15615	50.011	-0.738	727.00	0.05376	467.20	557.01
SDSS-II SN 09266	50.180	-1.002	154.00	0.03611	98.62	118.28
NGC 1309	50.527	-15.400	23.00	0.00713	14.62	17.75
NGC 1316	50.674	-37.208	15.70	0.00587	9.95	12.14
UGC 02708	50.954	40.558	59.40	0.02843	37.42	46.13
2MFGC 02845	51.795	46.943	91.50	0.01940	56.59	71.90
SDSS-II SN 16666	52.030	0.103	743.00	0.13294	457.13	585.73
SDSS-II SN 01588	52.149	-0.835	623.00	0.15050	382.28	491.92
FCCB 0602	52.366	-37.273	211.00	0.06300	128.84	167.10
SDSS-II SN 07778	52.449	-0.319	775.00	0.13870	472.33	614.43
SDSS-II SN 09967	53.186	-1.154	540.00	0.03046	323.58	432.31
NGC 1371	53.756	-24.933	16.40	0.00449	9.70	13.23
NGC 1380	54.115	-34.976	19.00	0.00626	11.14	15.39
SDSS-II SN 19407	54.219	1.199	851.00	0.08550	497.57	690.38
SDSS-II SN 16430	54.554	1.161	95.40	0.04059	55.33	77.72
SDSS-II SN 14269	54.759	0.279	976.00	0.28113	563.17	797.13
NSF J033923.74+350249.	54.850	35.046	23.30	0.00250	13.41	19.05
SDSS-II SN 15722	55.351	1.195	603.00	0.18250	342.84	496.06

SDSS-II SN 14777	55.362	1.177	787.00	0.09940	447.32	647.52
UGC 02829	55.462	8.160	63.80	0.02130	36.17	52.56
SDSS-II SN 17235	55.518	0.323	995.00	0.23375	563.32	820.18
SDSS-II SN 07980	55.744	0.257	180.00	0.16948	101.32	148.78
SDSS-II SN 14166	56.081	0.630	240.00	0.04105	133.93	199.16
NGC 1448	56.133	-44.645	15.90	0.00390	8.86	13.20
SDSS-II SN 19402	56.729	1.220	708.00	0.18150	388.40	591.95
SDSS-II SN 14175	56.939	0.960	866.00	0.17930	472.43	725.79
2MFGC 03182	57.430	-3.260	56.70	0.01285	30.52	47.78
SDSS J035012.90-001424	57.554	-0.240	789.00	0.27400	423.30	665.83
ESO 549- G 031	58.603	-19.190	97.90	0.02467	51.00	83.57
UGC 02998	64.143	2.759	49.30	0.01087	21.50	44.36
NGC 1559	64.399	-62.784	22.00	0.00435	9.51	19.84
UGC 03108	69.618	44.037	31.50	0.01321	10.97	29.53
UGC 03151	71.332	11.068	40.20	0.01460	12.87	38.09
NGC 1725	74.845	-11.132	52.60	0.01292	13.75	50.77
LSBG F119-024	74.862	-58.830	146.00	0.04500	38.13	140.93
UGC 03218	75.182	62.244	49.30	0.01743	12.61	47.66
NGC 1819	77.942	5.201	44.80	0.01491	9.36	43.81
ARP 327 NED04	80.457	6.677	113.00	0.03209	18.73	111.44
GALEXASC J052259.58-33	80.748	-33.466	310.00	0.11200	49.84	305.97
UGC 03329	84.136	16.649	39.30	0.01752	4.02	39.09
UGC 03341	85.481	18.492	72.30	0.01524	5.70	72.08
UGC 03365	88.148	66.820	97.50	0.01718	3.15	97.45
UGC 03375	88.856	51.911	48.70	0.01930	0.97	48.69
MCG -02-16-002	91.145	-12.625	23.30	0.00770	-0.47	23.30
UGC 03432	94.057	57.052	49.10	0.01667	-3.47	48.98
ESO 427- G 006	101.444	-31.231	38.70	0.00937	-7.68	37.93
NGC 2258	101.942	74.482	43.30	0.01354	-8.96	42.36
UGC 03576	103.279	50.034	62.40	0.01990	-14.33	60.73
CGCG 285-012	104.892	59.519	33.20	0.01110	-8.53	32.08
NGC 2320	106.425	50.581	50.80	0.01983	-14.36	48.73
UGC 03725	107.924	49.867	60.70	0.02058	-18.68	57.75
UGC 03845	111.678	47.094	32.00	0.01012	-11.82	29.74
UGC 04008 NED01	116.657	44.791	104.00	0.03088	-46.66	92.95
H-ZSNS J075039.32+1020	117.664	10.339	885.00	0.33960	-410.89	783.83
NGC 2441	117.978	73.016	38.10	0.01158	-17.87	33.65
CGCG 236-015 NED01	119.168	44.870	206.00	0.05024	-100.40	179.88
UGC 04195	121.279	66.783	62.90	0.01631	-32.66	53.76
ESO 561- G 018	122.313	-18.665	176.00	0.05056	-94.08	148.74
UGC 04322	125.007	62.831	78.60	0.02456	-45.09	64.38
[P94a] 081751.35+15532	125.169	15.729	460.00	0.08900	-264.96	376.03
H-ZSNS J082350.03+0328	125.958	3.481	670.00	0.18000	-393.42	542.33
SCP J082405.21+041122.	126.022	4.190	967.00	0.32000	-568.68	782.10
NGC 2595	126.925	21.479	47.80	0.01444	-28.72	38.21

PS1 J083612.60+440025.	129.053	44.007	601.00	0.14060	-378.65	466.72
NGC 2623	129.600	25.755	53.90	0.01846	-34.36	41.53
UGC 04614	132.318	36.120	84.20	0.02520	-56.69	62.26
ESO 125- G 006	132.649	-61.278	69.30	0.02233	-46.95	50.97
MCG -01-23-008	133.519	-7.183	86.30	0.03500	-59.43	62.58
MCG +08-17-043	137.387	50.282	49.70	0.01677	-36.58	33.65
NGC 2811	139.046	-16.313	21.30	0.00790	-16.09	13.96
MSACSS J092016.00+0033	140.067	0.561	436.00	0.15000	-334.32	279.87
NGC 2841	140.511	50.977	17.00	0.00213	-13.12	10.81
ESO 018- G 018	141.918	-80.180	62.30	0.01791	-49.04	38.43
NGC 2935	144.187	-21.128	24.70	0.00758	-20.03	14.45
NGC 2929	144.374	23.162	101.00	0.02505	-82.10	58.83
KK 1524	144.642	-63.973	62.60	0.01572	-51.05	36.23
2MFGC 07478	144.678	-12.335	258.00	0.05200	-210.51	149.17
ARP 321 NED01	144.723	-4.849	92.70	0.02211	-75.68	53.54
NGC 2962	145.225	5.166	23.60	0.00656	-19.38	13.46
NGC 2986	146.067	-21.278	29.70	0.00768	-24.64	16.58
NGC 3021	147.738	33.554	20.60	0.00514	-17.42	11.00
UGC 05378	150.133	4.407	46.30	0.01388	-40.15	23.06
CGCG 064-054	151.435	10.277	76.40	0.02510	-67.10	36.53
[GC96] J101020.28-1233	152.584	-12.553	922.00	0.43000	-818.45	424.53
MS 1008.1-1224:PPP 001	152.638	-12.618	863.00	0.30000	-766.45	396.64
NGC 3169	153.563	3.466	16.40	0.00413	-14.68	7.30
NGC 3147	154.224	73.401	30.10	0.00941	-27.10	13.09
NGC 3190	154.523	21.832	18.00	0.00424	-16.25	7.74
[PPG94] 101617.23+5107	154.861	50.872	875.00	0.42500	-792.12	371.72
UGC 05691	157.365	22.008	162.00	0.05500	-149.52	62.35
[MH93a] 103235.1-34110	158.717	-34.443	206.00	0.06900	-191.95	74.77
NGC 3294	159.068	37.325	32.50	0.00529	-30.36	11.61
NGC 3370	161.767	17.274	18.90	0.00427	-17.95	5.91
H-ZSNS J105356.22-0339	163.484	-3.653	978.00	0.36000	-937.65	278.02
H-ZSNS J105651.48-0358	164.214	-3.977	654.00	0.44000	-629.34	177.91
ARK 271	164.695	59.487	101.00	0.02313	-97.42	26.66
NGC 3644	170.387	2.811	114.00	0.02377	-112.40	19.04
ESO 570- G 020	170.921	-22.271	114.00	0.02850	-112.57	17.99
NGC 3663	171.000	-12.297	53.30	0.01674	-52.64	8.34
NGC 3672	171.260	-9.795	21.00	0.00621	-20.76	3.19
ESO 439- G 018	174.502	-32.325	91.50	0.02942	-91.08	8.77
UGC 06609	174.623	20.528	126.00	0.02572	-125.45	11.81
[PPG94] 114725.33+1059	177.499	10.716	960.00	0.35400	-959.09	41.89
NGC 3972	178.938	55.321	25.20	0.00284	-25.20	0.47
NGC 3982	179.117	55.125	24.60	0.00370	-24.60	0.38
EROS J115700.46-112638	179.252	-11.444	399.00	0.12000	-398.97	5.21
NGC 3987	179.337	25.195	40.90	0.01502	-40.90	0.47
NGC 4039	180.473	-18.886	15.20	0.00569	-15.20	-0.13

NGC 4061	181.006	20.232	67.90	0.02403	-67.89	-1.19
NGC 4070	181.047	20.410	109.00	0.02513	-108.98	-1.99
ESO 573- G 014	185.316	-21.996	68.90	0.02281	-68.60	-6.38
NGC 4493	187.785	0.614	73.10	0.02316	-72.43	-9.90
NGC 4520	188.458	-7.375	82.20	0.02544	-81.31	-12.09
NGC 4526	188.513	7.699	16.00	0.00149	-15.82	-2.37
NGC 4536	188.613	2.188	16.40	0.00603	-16.22	-2.46
IC 0807	190.552	-17.403	99.10	0.03770	-97.42	-18.15
IC 3690	190.705	10.357	86.90	0.02540	-85.39	-16.14
NGC 4639	190.718	13.257	26.80	0.00340	-26.33	-4.98
NGC 4680	191.728	-11.637	23.70	0.00831	-23.21	-4.82
NGC 4679	191.876	-39.571	45.20	0.01538	-44.23	-9.30
NGC 4704	192.193	41.921	72.20	0.02713	-70.57	-15.25
MCG -01-33-012	192.256	-9.457	64.30	0.01288	-62.83	-13.65
MRK 1337	193.145	-9.777	27.70	0.00853	-26.97	-6.30
IC 4042A	195.179	27.963	112.00	0.02791	-108.09	-29.32
IC 4062	195.244	39.859	115.00	0.03690	-110.95	-30.24
NGC 4948	196.234	-7.948	22.00	0.00375	-21.12	-6.15
2MASX J13050256+284420	196.261	28.739	66.40	0.02806	-63.74	-18.59
ESO 269- G 057	197.519	-46.437	42.90	0.01036	-40.91	-12.91
NGC 5005	197.734	37.059	14.10	0.00316	-13.43	-4.29
NGC 5018	198.254	-19.518	25.80	0.00939	-24.50	-8.08
CGCG 044-035	198.455	6.963	166.00	0.02396	-157.46	-52.55
ESO 576- G 017	198.803	-17.967	38.40	0.00900	-36.35	-12.38
KUG 1314+318B	199.213	31.581	137.00	0.02983	-129.37	-45.08
NGC 5061	199.521	-26.837	18.60	0.00689	-17.53	-6.22
IC 0883	200.147	34.140	80.60	0.02330	-75.67	-27.76
ESO 576- G 040	200.182	-22.051	20.70	0.00694	-19.43	-7.14
UGC 08399	200.439	31.237	112.00	0.02423	-104.95	-39.11
IC 4232	200.843	-26.109	81.40	0.03144	-76.07	-28.96
ESO 508- G 067	201.044	-23.877	72.30	0.02600	-67.48	-25.96
ESO 508- G 075	201.389	-24.652	103.00	0.03320	-95.91	-37.56
NGC 5177	202.351	11.797	100.00	0.02157	-92.49	-38.03
NGC 5185	202.509	13.416	95.40	0.02459	-88.13	-36.52
MCG -02-34-061	202.766	-15.101	56.20	0.01407	-51.82	-21.75
2MASX J13310895-331257	202.787	-33.216	157.00	0.05200	-144.75	-60.81
ESO 383- G 032	203.845	-34.160	71.80	0.02362	-65.67	-29.03
LCRS B133326.3-112212	204.023	-11.625	272.00	0.07429	-248.44	-110.73
NGC 5246	204.372	4.104	94.10	0.02312	-85.71	-38.83
NGC 5253	204.983	-31.640	3.67	0.00136	-3.33	-1.55
NGC 5283	205.274	67.672	39.30	0.01040	-35.54	-16.78
CGCG 190-050	206.396	35.611	107.00	0.02368	-95.84	-47.57
NGC 5308	206.752	60.973	21.20	0.00681	-18.93	-9.54
NGC 5304	207.506	-30.578	62.30	0.01329	-55.26	-28.77
MCG +08-25-047	207.704	49.315	93.80	0.02700	-83.05	-43.61

ESO 445- G 066	208.210	-30.710	95.70	0.02421	-84.33	-45.24
NGC 5440	210.754	34.758	53.60	0.01230	-46.06	-27.41
NGC 5468	211.645	-5.453	31.90	0.00948	-27.16	-16.74
NGC 5490	212.489	17.546	72.30	0.01620	-60.98	-38.84
[HSP2005] J141635.93+5	214.150	52.479	981.00	0.33700	-811.85	-550.69
SNLS 06D3gn HOST	214.436	52.361	818.00	0.25000	-674.65	-462.57
[NSB2006] J142150.02+5	215.458	53.137	794.00	0.22000	-646.74	-460.61
NGC 5584	215.599	-0.388	18.40	0.00546	-14.96	-10.71
IC 4423	216.574	26.246	106.00	0.03024	-85.13	-63.16
UGC 09391	218.654	59.338	26.10	0.00638	-20.38	-16.30
NGC 5698	219.311	38.454	61.00	0.01227	-47.20	-38.65
NGC 5728	220.599	-17.253	39.00	0.01010	-29.61	-25.38
MCG -01-38-002	222.153	-4.736	105.00	0.04250	-77.84	-70.47
CGCG 020-052	225.939	-3.303	93.60	0.02120	-65.09	-67.26
CGCG 193-015	226.136	35.965	87.30	0.00650	-60.49	-62.94
UGC 09758	227.867	13.484	97.90	0.02206	-65.68	-72.60
UGC 09798	229.174	7.397	122.00	0.03908	-79.76	-92.32
MCG -01-39-003	230.389	-7.448	21.00	0.00646	-13.39	-16.18
IC 1129	233.003	68.246	50.40	0.02190	-30.33	-40.25
MCG +11-19-025	238.232	65.936	116.00	0.03600	-61.07	-98.62
CGCG 108-013	239.140	20.052	100.00	0.03216	-51.29	-85.84
IC 1151	239.635	17.441	38.90	0.00724	-19.66	-33.56
NGC 6038	240.669	37.360	40.50	0.03133	-19.84	-35.31
NGC 6063	241.804	7.979	30.60	0.00950	-14.46	-26.97
UGC 10244	242.481	43.129	114.00	0.03253	-52.67	-101.10
UGC 10261	242.767	52.450	290.00	0.06343	-132.71	-257.85
ESO 584- G 007	243.222	-21.623	101.00	0.03166	-45.50	-90.17
NGC 6172	245.543	-1.515	76.70	0.01671	-31.75	-69.82
UGC 10432	247.663	41.211	159.00	0.03256	-60.43	-147.07
2MFGC 13321	249.372	40.880	104.00	0.02615	-36.64	-97.33
CGCG 168-029	250.805	32.678	230.00	0.05289	-75.62	-217.21
CGCG 224-104	252.463	40.434	81.50	0.02956	-24.56	-77.71
UGC 10704	255.455	79.038	182.00	0.06439	-45.71	-176.17
UGC 10738	257.774	5.852	68.60	0.02240	-14.53	-67.04
UGC 10743	257.878	7.995	34.70	0.00857	-7.29	-33.93
CGCG 111-016	258.893	16.325	89.30	0.03066	-17.20	-87.63
NGC 6411	263.887	60.813	54.50	0.01269	-5.80	-54.19
ARK 527	266.036	40.867	112.00	0.00600	-7.74	-111.73
NGC 6462	266.204	61.911	161.00	0.03841	-10.66	-160.65
NGC 6495	268.712	18.327	41.20	0.01043	-0.93	-41.19
UGC 11064	269.421	27.835	72.40	0.02349	-0.73	-72.40
CGCG 141-044	270.390	26.253	44.80	0.01550	0.31	-44.80
UGC 11097	270.604	26.041	69.40	0.01518	0.73	-69.40
NGC 6575	272.739	31.116	93.20	0.02332	4.45	-93.09
UGC 11149	272.791	49.865	61.30	0.05422	2.98	-61.23

UGC 11198	274.747	16.250	62.20	0.01507	5.15	-61.99
NGC 6627	275.662	15.698	49.90	0.01757	4.92	-49.66
UGC 11260	277.606	34.103	96.30	0.02063	12.75	-95.45
IC 4758	281.576	-65.757	43.00	0.01550	8.63	-42.13
IC 4830	288.452	-59.294	40.30	0.01464	12.76	-38.23
MCG +07-41-001	299.054	40.434	50.00	0.01588	24.28	-43.71
IC 4919	300.038	-55.373	37.10	0.01422	18.57	-32.12
CGCG 257-035	300.708	49.320	59.50	0.02290	30.38	-51.16
NGC 6916	305.888	58.344	33.80	0.01034	19.81	-27.38
NGC 6928	308.209	9.927	54.00	0.01570	33.40	-42.43
NGC 6930	308.245	9.875	61.80	0.01566	38.26	-48.54
NGC 6949	308.779	64.803	29.30	0.00870	18.35	-22.84
SDSS-II SN 01906	309.649	0.913	744.00	0.20023	474.73	-572.86
SDSS-II SN 01821	310.339	0.234	584.00	0.09610	378.03	-445.14
SDSS-II SN 14622	311.269	0.288	214.00	0.07112	141.15	-160.85
ESO 234- G 069	311.478	-51.391	41.10	0.01400	27.22	-30.79
SDSS J204809.20-010054	312.038	-1.015	861.00	0.21700	576.55	-639.46
SDSS-II SN 08477	312.040	-1.212	323.00	0.05634	216.30	-239.88
SDSS J204929.40+000016	312.373	0.005	242.00	0.03000	163.10	-178.78
SDSS-II SN 16624	312.712	-0.731	590.00	0.05710	400.20	-433.52
SDSS-II SN 01757	312.807	-0.958	654.00	0.12945	444.41	-479.81
SDSS-II SN 07547	312.923	0.860	345.00	0.10641	234.95	-252.63
SDSS-II SN 10922	312.956	0.930	800.00	0.14602	545.15	-585.50
SDSS-II SN 15199	313.224	0.069	436.00	0.41990	298.60	-317.70
SDSS-II SN 09126	313.671	-0.767	311.00	0.06950	214.75	-224.95
SDSS-II SN 13840	313.707	0.125	811.00	0.24190	560.37	-586.26
SDSS-II SN 05616	313.938	-0.591	592.00	0.10460	410.78	-426.29
SDSS-II SN 13744	314.060	-1.104	557.00	0.22254	387.34	-400.27
NGC 6986	314.128	-18.567	95.00	0.02873	66.14	-68.19
SDSS-II SN 16204	314.346	0.306	310.00	0.05454	216.69	-221.69
SDSS-II SN 02842	314.553	-0.324	214.00	0.05264	150.14	-152.50
SDSS-II SN 11102	315.159	-1.000	672.00	0.15392	476.49	-473.86
SDSS-II SN 06826	315.740	0.109	928.00	0.12322	664.62	-647.66
SDSS-II SN 14695	316.191	-0.451	722.00	0.16172	521.03	-499.81
SDSS-II SN 02387	316.214	0.632	633.00	0.04798	456.98	-438.01
SDSS-II SN 03546	316.645	0.045	803.00	0.19006	583.87	-551.27
SDSS-II SN 01825	316.820	0.359	757.00	0.15631	552.01	-518.01
SDSS-II SN 07949	317.838	0.358	791.00	0.06990	586.33	-530.94
2MASX J21124822+005505	318.201	0.918	590.00	0.14585	439.84	-393.25
SDSS-II SN 03430	318.274	0.760	761.00	0.09763	567.96	-506.50
SDSS-II SN 06371	318.407	-0.991	932.00	0.12079	697.03	-618.69
SDSS-II SN 03057	318.491	0.350	684.00	0.13579	512.21	-453.32
SDSS-II SN 02388	318.596	0.461	890.00	0.06004	667.56	-588.61
SDSS-II SN 01725	318.619	0.770	267.00	0.13648	200.34	-176.50
SDSS-II SN 13333	318.649	0.278	956.00	0.27733	717.65	-631.60

SDSS-II SN 15192	318.767	0.451	702.00	0.13955	527.93	-462.71
SDSS-II SN 16105	319.003	-1.109	813.00	0.13964	613.61	-533.35
SDSS J211735.50+001856	319.398	0.316	999.00	0.25500	758.49	-650.15
SDSS-II SN 14648	319.656	-1.055	429.00	0.06133	326.97	-277.72
SDSS-II SN 10039	319.827	1.183	435.00	0.13430	332.38	-280.62
SDSS-II SN 15467 HOST	320.020	-0.177	775.00	0.21000	593.86	-497.95
UGC 11723	320.073	-1.684	60.10	0.01634	46.09	-38.57
UGC 11725	320.257	9.178	106.00	0.02069	81.51	-67.77
SDSS-II SN 13730	320.292	0.933	892.00	0.25650	686.22	-569.88
SDSS-II SN 15830	320.347	0.627	717.00	0.13922	552.04	-457.54
SDSS-II SN 19908	320.476	-0.287	855.00	0.23555	659.51	-544.12
SDSS-II SN 12578	320.514	0.518	563.00	0.13504	434.51	-358.01
SDSS-II SN 06509	320.863	-1.192	524.00	0.05820	406.43	-330.74
SDSS J212342.94-005036	320.929	-0.843	181.00	0.06000	140.52	-114.08
SDSS-II SN 15396	321.008	-0.366	386.00	0.06193	300.01	-242.88
SDSS-II SN 06885	321.144	0.979	507.00	0.04946	394.82	-318.07
SDSS-II SN 05611	322.093	-0.098	420.00	0.05028	331.38	-258.04
SDSS-II SN 04910	322.124	-0.171	957.00	0.13780	755.40	-587.55
SDSS-II SN 08363	322.225	1.079	408.00	0.08826	322.49	-249.92
SDSS-II SN 16452	322.557	-0.282	815.00	0.20611	647.08	-495.50
UGC 11758	322.740	13.986	101.00	0.02760	80.39	-61.15
SDSS-II SN 14985	323.096	-0.043	653.00	0.08865	522.17	-392.11
SDSS-II SN 11083	323.479	0.180	768.00	0.12193	617.20	-457.05
SDSS-II SN 09292	323.557	0.756	637.00	0.16678	512.44	-378.39
SDSS-II SN 05626	323.603	-0.603	480.00	0.08582	386.36	-284.82
SDSS-II SN 06472	323.762	-0.171	549.00	0.05090	442.81	-324.54
SDSS-II SN 13998	324.076	-0.026	286.00	0.12268	231.60	-167.80
SDSS-II SN 01615	324.653	-0.947	687.00	0.08267	560.36	-397.45
SDSS-II SN 13300	324.844	-0.541	414.00	0.09142	338.48	-238.38
SDSS-II SN 02116	324.895	0.055	924.00	0.30780	755.92	-531.37
SDSS J214000.48-000002	325.002	-0.001	430.00	0.14420	352.24	-246.63
SDSS-II SN 15347	325.405	1.204	931.00	0.27001	766.39	-528.59
SDSS-II SN 16393	326.501	-0.061	920.00	0.31643	767.18	-507.77
SDSS-II SN 20152	326.555	1.028	872.00	0.29485	727.61	-480.59
NGC 7131	326.900	-13.183	52.70	0.01807	44.15	-28.78
SDSS J215108.63-005033	327.786	-0.843	880.00	0.31000	744.54	-469.11
SDSS-II SN 13721	327.876	0.511	566.00	0.28230	479.35	-300.97
SDSS-II SN 03040	328.911	-1.074	951.00	0.21539	814.40	-491.07
SDSS-II SN 21846	328.974	-0.049	514.00	0.06864	440.46	-264.93
SDSS-II SN 17393	329.378	-0.929	905.00	0.21740	778.79	-460.99
SDSS-II SN 02502	329.398	-0.839	315.00	0.21853	271.13	-160.36
SDSS-II SN 05630	329.778	-0.559	388.00	0.13169	335.26	-195.30
SDSS-II SN 09477	329.880	1.222	339.00	0.10090	293.23	-170.11
SDSS J215947.23-004359	329.947	-0.733	863.00	0.28000	746.98	-432.19
SDSS-II SN 14224	330.076	-0.532	678.00	0.12818	587.61	-338.22

SDSS-II SN 09377	330.104	-0.578	216.00	0.12610	187.26	-107.66
SDSS-II SN 01305	330.130	0.595	771.00	0.09786	668.58	-383.98
SDSS-II SN 00694	330.155	-0.624	495.00	0.12749	429.35	-246.34
SDSS-II SN 16312	330.623	0.347	1000.00	0.28907	871.41	-490.56
SDSS-II SN 12888	330.948	0.423	732.00	0.20110	639.90	-355.46
SDSS-II SN 01902	330.961	-0.795	881.00	0.21673	770.25	-427.63
SDSS-II SN 11002	331.008	0.585	395.00	0.04926	345.50	-191.45
SDSS-II SN 13018	331.218	0.526	610.00	0.14120	534.64	-293.70
SDSS-II SN 05490	331.290	-0.806	338.00	0.09807	296.45	-162.37
SDSS-II SN 21058	331.576	-0.076	631.00	0.16430	554.93	-300.35
SDSS-II SN 19893	331.945	-0.703	499.00	0.10939	440.37	-234.69
SDSS-II SN 21808	332.065	0.631	795.00	0.09415	702.37	-372.43
SDSS-II SN 06169	332.187	-0.986	730.00	0.08738	645.67	-340.60
SDSS-II SN 03166	332.444	-0.640	763.00	0.08896	676.45	-352.97
NGC 7222	332.716	2.106	174.00	0.04145	154.64	-79.76
SDSS-II SN 06171	332.911	-0.924	816.00	0.08114	726.49	-371.58
SDSS-II SN 08196	333.043	-0.913	645.00	0.08002	574.92	-292.39
SNLS 06D4dh HOST	333.582	-17.585	837.00	0.30300	749.59	-372.39
SDSS-II SN 07496	333.610	-0.781	990.00	0.55134	886.83	-440.03
SDSS-II SN 06800	333.923	-1.123	605.00	0.04775	543.41	-265.94
SDSS-II SN 10697	333.926	-0.603	382.00	0.09911	343.12	-167.90
IC 5179	334.038	-36.844	41.60	0.01141	37.40	-18.21
SDSS-II SN 07798	334.196	-0.183	477.00	0.11867	429.44	-207.63
UGC 11977	334.389	33.256	161.00	0.03764	145.18	-69.59
NGC 7250	334.574	40.563	19.70	0.00389	17.79	-8.46
SDSS-II SN 15627	334.761	-0.788	525.00	0.09771	474.88	-223.86
CGCG 473-011	334.776	24.598	115.00	0.02754	104.03	-49.01
SDSS J221908.85+131040	334.787	13.178	117.00	0.02782	105.85	-49.84
SDSS-II SN 01981	334.859	0.928	580.00	0.05683	525.05	-246.41
SDSS-II SN 08382	335.168	1.230	538.00	0.08704	488.26	-225.94
UGC 11994	335.222	33.295	72.50	0.01625	65.83	-30.39
SDSS-II SN 17924	336.041	0.326	471.00	0.14572	430.42	-191.27
SDSS-II SN 01136	336.201	1.009	760.00	0.11268	695.37	-306.68
SDSS-II SN 02092	336.456	-0.239	592.00	0.14339	542.72	-236.48
SDSS-II SN 10324	336.618	0.587	896.00	0.25172	822.42	-355.59
SDSS-II SN 13074	336.939	0.203	728.00	0.19412	669.82	-285.17
SDSS-II SN 03065	336.978	-1.155	908.00	0.15577	835.68	-355.11
SDSS J222805.78-010731	337.024	-1.125	613.00	0.24400	564.37	-239.28
SDSS-II SN 21693	337.063	1.003	403.00	0.08915	371.14	-157.05
SDSS-II SN 19231	337.142	0.740	611.00	0.08557	563.02	-237.34
SDSS-II SN 14177	337.592	0.516	919.00	0.29370	849.61	-350.32
SDSS-II SN 13035	337.612	0.969	845.00	0.03092	781.31	-321.84
SDSS-II SN 08387	337.680	1.239	547.00	0.13181	506.02	-207.74
SDSS-II SN 05513	337.797	0.118	965.00	0.27777	893.44	-364.67
SDSS-II SN 01679	337.800	-1.020	837.00	0.12937	774.96	-316.25

SDSS-II SN 09317	338.074	0.643	953.00	0.25609	884.07	-355.86
SDSS-II SN 13307	338.121	-0.544	751.00	0.08853	696.91	-279.86
SDSS-II SN 06180	338.221	-0.973	605.00	0.05705	561.82	-224.47
SDSS-II SN 12327	338.340	1.133	842.00	0.07381	782.54	-310.79
CGCG 452-024	338.460	20.805	163.00	0.03445	151.62	-59.84
SDSS-II SN 07037	338.519	0.141	474.00	0.09017	441.08	-173.57
SDSS-II SN 19038	338.870	0.004	791.00	0.19168	737.82	-285.14
SDSS-II SN 16727	339.005	0.132	1000.00	0.26074	933.61	-358.29
SDSS J223602.72-002204	339.015	-0.368	759.00	0.25800	708.66	-271.81
SDSS J223618.96+002846	339.079	0.480	980.00	0.23800	915.39	-349.94
SDSS-II SN 15202	339.137	0.117	944.00	0.13248	882.11	-336.18
SDSS-II SN 15391	339.624	-1.171	586.00	0.08816	549.33	-204.03
SDSS-II SN 03812	339.656	0.104	152.00	0.06039	142.52	-52.84
SDSS-II SN 02681	339.750	-0.632	941.00	0.23986	882.84	-325.70
SDSS-II SN 21813	339.772	1.185	950.00	0.32517	891.41	-328.47
SDSS-II SN 07254	339.831	0.543	794.00	0.36624	745.31	-273.76
UGC 12133	339.884	8.613	58.20	0.02476	54.65	-20.02
SDSS-II SN 11675	340.068	-0.482	941.00	0.08441	884.63	-320.79
NGC 7329	340.101	-66.479	44.20	0.01085	41.56	-15.04
SDSS-II SN 13431	340.832	0.459	411.00	0.05751	388.21	-134.95
SDSS-II SN 21185	340.844	-0.717	938.00	0.23394	886.06	-307.80
SDSS-II SN 11022	340.951	-1.216	683.00	0.13160	645.60	-222.91
SDSS-II SN 07040	341.108	0.147	407.00	0.12752	385.08	-131.78
SDSS-II SN 17973	341.134	0.194	569.00	0.14800	538.43	-183.99
SDSS-II SN 05151	341.147	-1.031	771.00	0.15909	729.64	-249.14
SDSS-II SN 01683	341.170	-0.865	821.00	0.08496	777.06	-264.98
SDSS-II SN 15601	341.172	0.499	714.00	0.10650	675.79	-230.43
SDSS-II SN 16877	341.205	1.164	942.00	0.12927	891.77	-303.49
2MFGC 17122	341.392	-8.751	128.00	0.04990	121.31	-40.84
SDSS-II SN 01243	341.660	-0.765	976.00	0.19843	926.42	-307.10
SDSS-II SN 21697	342.025	0.878	519.00	0.14687	493.67	-160.16
SDSS-II SN 05039	342.326	-0.039	693.00	0.08271	660.29	-210.39
SDSS-II SN 20648	342.380	0.731	936.00	0.09534	892.09	-283.32
SDSS-II SN 11914	342.457	1.190	716.00	0.11049	682.70	-215.82
SDSS-II SN 16055	342.487	1.140	622.00	0.10145	593.17	-187.17
SDSS-II SN 16606	342.611	1.223	951.00	0.22560	907.54	-284.21
SDSS-II SN 20052	343.077	1.234	940.00	0.15766	899.30	-273.62
SDSS-II SN 10121	343.205	1.201	354.00	0.01572	338.90	-102.29
SDSS-II SN 11100	343.301	1.172	771.00	0.21005	738.49	-221.54
SDSS-II SN 02645	343.461	0.717	654.00	0.09111	626.94	-186.17
SDSS-II SN 09853	343.581	-0.379	570.00	0.06773	546.76	-161.11
SDSS-II SN 10703	343.645	-0.571	232.00	0.06795	222.61	-65.33
SDSS-II SN 10853	343.775	0.744	458.00	0.06667	439.76	-127.97
SDSS-II SN 07439	344.095	-0.283	515.00	0.07880	495.28	-141.13
SDSS-II SN 17682	344.108	0.502	861.00	0.30481	828.09	-235.77

SDSS-II SN 13058	344.370	-0.263	733.00	0.18710	705.89	-197.49
IC 5270	344.479	-35.858	24.60	0.00662	23.70	-6.58
SDSS-II SN 03729	344.512	0.605	569.00	0.25429	548.34	-151.95
SDSS-II SN 06520	344.953	0.186	627.00	0.10907	605.50	-162.77
NGC 7448	345.015	15.980	31.20	0.00732	30.14	-8.07
SDSS-II SN 10460	345.029	-1.204	932.00	0.22034	900.37	-240.76
SDSS-II SN 19197	345.042	-0.092	662.00	0.17971	639.57	-170.86
SDSS-II SN 13641 HOST	345.217	-0.982	797.00	0.21000	770.62	-203.35
SDSS J230248.35-005235	345.701	-0.877	901.00	0.27010	873.09	-222.53
SDSS-II SN 18475	345.703	0.049	675.00	0.18164	654.09	-166.69
NGC 7469	345.815	8.874	66.40	0.01632	64.38	-16.27
SDSS-II SN 08947	345.841	1.182	731.00	0.18136	708.79	-178.82
SDSS-II SN 03549	345.953	0.777	710.00	0.41244	688.77	-172.32
APMUKS(BJ) B230147.96-	346.143	-37.342	319.00	0.10100	309.72	-76.40
IC 1468	346.281	-3.204	143.00	0.03207	138.92	-33.91
SDSS-II SN 18212	346.348	0.901	567.00	0.18423	550.98	-133.82
SDSS-II SN 21732	346.459	0.416	807.00	0.06225	784.57	-188.95
SDSS-II SN 21807	347.334	0.757	444.00	0.03240	433.20	-97.35
SDSS-II SN 08867	347.480	0.364	786.00	0.10367	767.31	-170.39
SDSS-II SN 01741	347.723	-0.568	828.00	0.20496	809.07	-176.06
SDSS-II SN 13442	347.756	-0.314	323.00	0.10244	315.65	-68.50
SDSS-II SN 19652	347.875	1.024	875.00	0.24595	855.48	-183.80
NGC 7541	348.683	4.534	24.60	0.00897	24.12	-4.83
SDSS-II SN 10288	348.757	0.072	562.00	0.05122	551.21	-109.58
ESO 291- G 011	348.800	-44.737	127.00	0.04300	124.58	-24.67
NGC 7549	348.822	19.042	68.10	0.01647	66.81	-13.20
SDSS-II SN 19657	348.992	-0.381	801.00	0.33776	786.26	-152.95
SDSS-II SN 05671	349.201	-0.574	286.00	0.15156	280.93	-53.59
SDSS-II SN 16218	349.561	0.766	515.00	0.02966	506.48	-93.31
SDSS-II SN 19432	349.751	0.131	961.00	0.28223	945.67	-170.99
SDSS-II SN 02568	349.872	0.600	848.00	0.11821	834.79	-149.11
SDSS-II SN 20874	350.232	0.508	785.00	0.13064	773.62	-133.18
UGC 12538	350.291	33.400	48.00	0.01633	47.31	-8.09
2MASX J23211915+001532	350.330	0.259	316.00	0.03078	311.51	-53.08
SDSS-II SN 15210	350.353	0.928	699.00	0.12596	689.12	-117.14
SDSS-II SN 00859	350.552	0.387	981.00	0.27830	967.69	-161.04
SDSS-II SN 06415	350.582	-0.078	580.00	0.18373	572.18	-94.91
SDSS-II SN 08644	351.024	0.456	1000.00	0.11832	987.75	-156.02
SDSS-II SN 00692	351.071	-0.946	458.00	0.19728	452.45	-71.09
SDSS-II SN 18643	351.200	0.944	515.00	0.11822	508.94	-78.79
SDSS-II SN 21409	351.397	-0.715	939.00	0.19769	928.44	-140.45
SDSS-II SN 06921	351.502	1.016	487.00	0.14211	481.65	-71.96
ESSENCE J232704.38-083	351.768	-8.646	705.00	0.30000	697.74	-100.94
ESSENCE n278 HOST	352.073	-9.387	822.00	0.30400	814.15	-113.36
SDSS-II SN 13445	352.190	-0.338	959.00	0.22323	950.11	-130.31

SDSS-II SN 20492	352.264	-1.190	699.00	0.09600	692.64	-94.09
SDSS-II SN 17421	352.386	0.677	962.00	0.27810	953.52	-127.47
ESSENCE m032 HOST	352.397	-9.980	494.00	0.15500	489.66	-65.36
ESSENCE m043 HOST	352.466	-8.946	666.00	0.26600	660.25	-87.33
SDSS-II SN 10714	353.170	-0.563	381.00	0.05760	378.30	-45.31
SDSS-II SN 16071	353.435	-0.927	186.00	0.09274	184.78	-21.26
SDSS-II SN 01650	353.510	0.684	945.00	0.30434	938.94	-106.81
SDSS-II SN 00717	353.628	0.766	469.00	0.13039	466.10	-52.05
SDSS-II SN 20893	353.809	0.943	674.00	0.08462	670.07	-72.69
SDSS-II SN 17760	354.082	0.064	705.00	0.19692	701.24	-72.69
SDSS-II SN 00718	354.211	0.678	947.00	0.16164	942.17	-95.52
SDSS-II SN 10327	354.376	1.053	879.00	0.08488	874.77	-86.15
SDSS-II SN 01533	354.457	-0.802	536.00	0.14447	533.49	-51.78
SDSS-II SN 15241	354.655	-0.470	876.00	0.03565	872.19	-81.61
SDSS-II SN 18610	354.766	0.414	941.00	0.12911	937.08	-85.84
SDSS-II SN 01354	354.803	0.090	925.00	0.24944	921.20	-83.79
SDSS-II SN 04393	354.990	0.135	375.00	0.06019	373.57	-32.75
SDSS-II SN 13468	355.044	0.174	802.00	0.05979	799.00	-69.29
SDSS-II SN 13180	355.399	0.847	786.00	0.25101	783.47	-63.05
ARP 295A	355.447	-3.667	93.10	0.02202	92.81	-7.39
SDSS-II SN 15182	355.826	-1.157	888.00	0.14298	885.64	-64.64
SDSS-II SN 15603	355.942	-1.192	968.00	0.26259	965.57	-68.49
SDSS-II SN 17842	356.013	-0.938	581.00	0.12492	579.59	-40.40
SDSS-II SN 01812	356.322	0.304	770.00	0.45978	768.41	-49.40
SDSS-II SN 13893	356.395	-0.091	749.00	0.28287	747.52	-47.09
SDSS-II SN 15242	356.752	-0.447	354.00	0.07117	353.43	-20.06
CGCG 498-012	356.852	28.393	130.00	0.03013	129.80	-7.14
SDSS-II SN 10727	356.942	-0.496	486.00	0.07132	485.31	-25.92
SDSS-II SN 21871	357.094	-0.582	855.00	0.20270	853.90	-43.34
SDSS-II SN 17633	357.393	-1.173	658.00	0.22861	657.32	-29.92
SDSS-II SN 03757	357.401	0.432	464.00	0.07036	463.52	-21.04
SDSS-II SN 12876	357.752	-0.009	686.00	0.10019	685.47	-26.91
SDSS-II SN 19129	357.825	-0.634	442.00	0.10126	441.68	-16.78
NGC 7761	357.870	-13.382	88.30	0.02398	88.24	-3.28
SDSS-II SN 16150	357.947	0.973	847.00	0.07171	846.46	-30.34
ESO 471- G 027	357.960	-27.965	81.20	0.02936	81.15	-2.89
SDSS-II SN 16406	358.086	0.361	965.00	0.17364	964.46	-32.23
SDSS-II SN 10612	358.236	0.199	659.00	0.16676	658.69	-20.29
SDSS-II SN 06280	358.318	1.155	548.00	0.18775	547.76	-16.09
SDSS-II SN 15160	358.429	-0.579	922.00	0.25223	921.65	-25.27
SDSS-II SN 12895	358.621	-0.144	910.00	0.12770	909.74	-21.91
SDSS-II SN 03868	359.126	0.100	946.00	0.19607	945.89	-14.44
SDSS J235721.96+000621	359.342	0.106	807.00	0.19610	806.95	-9.27
SDSS-II SN 12825	359.544	-1.182	607.00	0.15012	606.98	-4.83
MCG -01-01-010	359.739	-2.250	106.00	0.02550	106.00	-0.48

```
# ================================================================================
# INFINITE BOX THEORY: DEEP SPACE FLUID RADAR (1000 Mpc)
# Mapping Macro-Currents (Headwinds & Tailwinds) via Contour Interpolation
# ================================================================================

import pandas as pd
import numpy as np
from scipy.interpolate import griddata
import plotly.graph_objects as go
import warnings
warnings.filterwarnings('ignore')

print("1. Initializing Deep Space Hydrodynamic Radar (1000 Mpc)...")

# ------------------------------------------------------------------------------
# A. LOAD & PREP LOCALIZED DATA
# ------------------------------------------------------------------------------
col_names = [
    "FRN", "Exclusion_Code", "D", "Galaxy_Name", "Distance_Modulus", "Error",
    "Distance_Mpc", "Method", "REFCODE", "Target_ID", "Redshift_z",
    "Hubble_const", "Blank", "Adopted_LMC", "Notes"
]

# Load NED Data
df_ned = pd.read_csv("/NED30.5.1-D-17.1.2-20200415.csv", skiprows=13, names=col_names, low_memory=False)

for col in ['Redshift_z', 'Distance_Mpc', 'Error']:
    df_ned[col] = pd.to_numeric(df_ned[col], errors='coerce')

df_ned = df_ned.dropna(subset=['Redshift_z', 'Distance_Mpc', 'Galaxy_Name', 'Error'])

# Pristine SNIa Filter (< 5% error) & Expanded 1000 Mpc Boundary
df_sc = df_ned[df_ned['Method'].str.contains('SNIa', na=False, case=False)].copy()
df_sc['Error_Margin'] = (df_sc['Error'] / df_sc['Distance_Mpc']) * 100
```

```
df_local = df_sc[(df_sc['Error_Margin'] <= 5.0) & (df_sc['Distance_Mpc'] <= 1000.0)]

# Load Roleup Data for RA/Dec
df_roleup = pd.read_csv("/Roleup.csv", low_memory=False)[['Object Name', 'RA', 'Dec']].dropna()
df_unique = pd.merge(df_local, df_roleup, left_on='Galaxy_Name', right_on='Object Name',
how='inner').drop_duplicates(subset=['Galaxy_Name']).copy()

# Convert to 2D Cartesian Coordinates
df_unique['X'] = df_unique['Distance_Mpc'] * np.cos(np.radians(df_unique['RA']))
df_unique['Y'] = df_unique['Distance_Mpc'] * np.sin(np.radians(df_unique['RA']))

print(f"   -> Radar locked on {len(df_unique)} deep-space structural nodes.")

# ---------------------------------------------------------------------------
# B. CALCULATE FLUID DELTAS (The Barometric Pressure)
# ---------------------------------------------------------------------------
print("2. Calculating True Fluid Eddies for all nodes...")

H0 = 68.1
c = 299792.458

# Expected Z if the universe was an empty, uniform vacuum
df_unique['Expected_Z'] = (df_unique['Distance_Mpc'] * H0) / c

# The Delta: Actual Redshift vs Expected Redshift
# Positive Delta = Headwind (Red), Negative Delta = Tailwind (Blue)
df_unique['Fluid_Delta'] = df_unique['Redshift_z'] - df_unique['Expected_Z']

# ---------------------------------------------------------------------------
# C. INTERPOLATE THE WEATHER MAP
# ---------------------------------------------------------------------------
print("3. Interpolating Macro-Currents (Cosmic Wind)...")

# Create a dense 2D grid (Expanded resolution for 1000 Mpc scale)
```

```
grid_x, grid_y = np.mgrid[-1000:1000:400j, -1000:1000:400j]

# Interpolate the Fluid_Delta values across the empty space between galaxies
grid_z = griddata(
    (df_unique['X'], df_unique['Y']),
    df_unique['Fluid_Delta'],
    (grid_x, grid_y),
    method='cubic' # Creates smooth, sweeping weather patterns
)

# -----------------------------------------------------------------------------
# D. RENDER THE METEOROLOGICAL DASHBOARD
# -----------------------------------------------------------------------------
fig = go.Figure()

# 1. The Weather Radar Overlay (Contour Map)
fig.add_trace(go.Contour(
    x=grid_x[:,0], y=grid_y[0,:], z=grid_z.T,
    colorscale='RdBu_r',
    zmin=-0.05, zmax=0.05, # Kept the same clamp to highlight severe gradients at scale
    opacity=0.6,
    colorbar=dict(title="Fluid Eddy (ΔZ)"),
    name='Macro-Current (Fluid Drag)'
))

# 2. Plot the Rendered Actuals (The Galaxies)
fig.add_trace(go.Scatter(
    x=df_unique['X'], y=df_unique['Y'],
    mode='markers',
    marker=dict(size=3, color='white', opacity=0.9, line=dict(width=0.5, color='black')),
    name='Historical Sediment (Sensor Nodes)',
    text=df_unique['Galaxy_Name'],
    hovertemplate="<b>%{text}</b><br>Fluid Delta: %{customdata:.4f}<extra></extra>",
    customdata=df_unique['Fluid_Delta']
```

```
))

# 3. Plot Earth / Origin (The Observer)
fig.add_trace(go.Scatter(
    x=[0], y=[0],
    mode='markers',
    marker=dict(size=12, color='cyan', symbol='cross', line=dict(width=2, color='white')),
    name='Earth (Anchor Zone)'
))

# Format Geometry
fig.update_layout(
    title='IBT Deep Space Fluid Radar: Macro-Current Weather Map (1000 Mpc)',
    xaxis_title='X Axis (Mpc)',
    yaxis_title='Y Axis (Mpc)',
    xaxis=dict(range=[-1100, 1100], showgrid=True, gridcolor='#333', zerolinecolor='white'),
    yaxis=dict(range=[-1100, 1100], showgrid=True, gridcolor='#333', zerolinecolor='white', scaleanchor="x", scaleratio=1),
    plot_bgcolor='black',
    paper_bgcolor='black',
    font=dict(color='white'),
    margin=dict(l=40, r=40, b=40, t=60)
)

fig.show()
print("Deep Space Radar is live. Scanning 1000 Mpc radius.")
```

```
# ================================================================================
# INFINITE BOX THEORY: DEEP SPACE VECTOR FLUID RADAR (1000 Mpc)
# Mapping Contour Pressures & Overlaying the Cosmic Wind (Normalized Quiver)
# ================================================================================

import pandas as pd
import numpy as np
from scipy.interpolate import griddata
import plotly.graph_objects as go
import plotly.figure_factory as ff
import warnings
warnings.filterwarnings('ignore')

print("1. Initializing Deep Space Hydrodynamic Radar with Vector Telemetry...")

# -----------------------------------------------------------------------------
# A. LOAD & PREP LOCALIZED DATA
# -----------------------------------------------------------------------------
col_names = [
    "FRN", "Exclusion_Code", "D", "Galaxy_Name", "Distance_Modulus", "Error",
    "Distance_Mpc", "Method", "REFCODE", "Target_ID", "Redshift_z",
    "Hubble_const", "Blank", "Adopted_LMC", "Notes"
]

df_ned = pd.read_csv("/NED30.5.1-D-17.1.2-20200415.csv", skiprows=13, names=col_names, low_memory=False)

for col in ['Redshift_z', 'Distance_Mpc', 'Error']:
    df_ned[col] = pd.to_numeric(df_ned[col], errors='coerce')

df_ned = df_ned.dropna(subset=['Redshift_z', 'Distance_Mpc', 'Galaxy_Name', 'Error'])

# Pristine SNIa Filter (< 5% error) & Expanded 1000 Mpc Boundary
df_sc = df_ned[df_ned['Method'].str.contains('SNIa', na=False, case=False)].copy()
df_sc['Error_Margin'] = (df_sc['Error'] / df_sc['Distance_Mpc']) * 100
```

```
df_local = df_sc[(df_sc['Error_Margin'] <= 5.0) & (df_sc['Distance_Mpc'] <= 1000.0)]

df_roleup = pd.read_csv("/Roleup.csv", low_memory=False)[['Object Name', 'RA', 'Dec']].dropna()
df_unique = pd.merge(df_local, df_roleup, left_on='Galaxy_Name', right_on='Object Name',
how='inner').drop_duplicates(subset=['Galaxy_Name']).copy()

df_unique['X'] = df_unique['Distance_Mpc'] * np.cos(np.radians(df_unique['RA']))
df_unique['Y'] = df_unique['Distance_Mpc'] * np.sin(np.radians(df_unique['RA']))

print(f"   -> Radar locked on {len(df_unique)} deep-space structural nodes.")

# -----------------------------------------------------------------------------
# B. CALCULATE FLUID DELTAS
# -----------------------------------------------------------------------------
print("2. Calculating True Fluid Eddies for all nodes...")

H0 = 68.1
c = 299792.458
df_unique['Expected_Z'] = (df_unique['Distance_Mpc'] * H0) / c
df_unique['Fluid_Delta'] = df_unique['Redshift_z'] - df_unique['Expected_Z']

# -----------------------------------------------------------------------------
# C. INTERPOLATE WEATHER & NORMALIZE VECTOR WINDS
# -----------------------------------------------------------------------------
print("3. Interpolating Macro-Currents & Calculating Vector Winds...")

grid_x, grid_y = np.mgrid[-1000:1000:400j, -1000:1000:400j]

grid_z = griddata(
   (df_unique['X'], df_unique['Y']),
   df_unique['Fluid_Delta'],
   (grid_x, grid_y),
   method='cubic'
)
```

```
# Calculate Gradient (Slope)
dy, dx = np.gradient(grid_z.T)
u_wind = -dx
v_wind = -dy

# Down-sample the grid tightly
skip = 8
x_q = grid_x[::skip, ::skip].flatten()
y_q = grid_y[::skip, ::skip].flatten()
u_q = u_wind[::skip, ::skip].flatten()
v_q = v_wind[::skip, ::skip].flatten()

# Filter out the invalid NaN space so Plotly doesn't crash
valid = ~np.isnan(u_q) & ~np.isnan(v_q)
x_valid = x_q[valid]
y_valid = y_q[valid]
u_valid = u_q[valid]
v_valid = v_q[valid]

# VECTOR NORMALIZATION: Force all arrows to be exactly 30 Mpc long
arrow_length = 30.0
magnitude = np.sqrt(u_valid**2 + v_valid**2)
magnitude[magnitude == 0] = 1.0 # Prevent division by zero

u_norm = (u_valid / magnitude) * arrow_length
v_norm = (v_valid / magnitude) * arrow_length

# ---------------------------------------------------------------------------
# D. RENDER THE DASHBOARD
# ---------------------------------------------------------------------------
fig = go.Figure()

# 1. The Weather Radar Overlay (Contour Map)
```

```
fig.add_trace(go.Contour(
    x=grid_x[:,0], y=grid_y[0,:], z=grid_z.T,
    colorscale='RdBu_r',
    zmin=-0.05, zmax=0.05,
    opacity=0.6,
    colorbar=dict(title="Fluid Eddy (ΔZ)"),
    name='Macro-Current (Fluid Drag)'
))

# 2. Add the Quiver Plot (Normalized Arrows)
fig_quiver = ff.create_quiver(
    x_valid, y_valid, u_norm, v_norm,
    scale=1, # Scale is strictly 1 because length is baked into the math
    arrow_scale=0.3,
    line=dict(color='cyan', width=2),
    name='Cosmic Wind Vectors'
)
fig.add_traces(fig_quiver.data)

# 3. Plot the Galaxies
fig.add_trace(go.Scatter(
    x=df_unique['X'], y=df_unique['Y'],
    mode='markers',
    marker=dict(size=3, color='white', opacity=0.9, line=dict(width=0.5, color='black')),
    name='Historical Sediment',
    text=df_unique['Galaxy_Name'],
    hovertemplate="<b>%{text}</b><br>Fluid Delta: %{customdata:.4f}<extra></extra>",
    customdata=df_unique['Fluid_Delta']
))

# 4. Plot Earth
fig.add_trace(go.Scatter(
    x=[0], y=[0],
    mode='markers',
```

```
    marker=dict(size=12, color='lime', symbol='cross', line=dict(width=2, color='white')),
    name='Earth (Anchor Zone)'
))

fig.update_layout(
    title='IBT Deep Space Fluid Radar: Vector Flow Topography (1000 Mpc)',
    xaxis_title='X Axis (Mpc)',
    yaxis_title='Y Axis (Mpc)',
    xaxis=dict(range=[-1100, 1100], showgrid=True, gridcolor='#333', zerolinecolor='white'),
    yaxis=dict(range=[-1100, 1100], showgrid=True, gridcolor='#333', zerolinecolor='white', scaleanchor="x", scaleratio=1),
    plot_bgcolor='black',
    paper_bgcolor='black',
    font=dict(color='white'),
    margin=dict(l=40, r=40, b=40, t=60)
)

fig.show()
print("Vector Flow actively mapped. Cyan arrows locked to 30 Mpc rendering scale.")
```

```
# ================================================================================
# INFINITE BOX THEORY: DEEP SPACE VECTOR FLUID RADAR (1000 Mpc)
# Mapping Contour Pressures, Cosmic Wind, & Physical Sediment Density (KDE)
# ================================================================================

import pandas as pd
import numpy as np
from scipy.interpolate import griddata
from scipy.stats import gaussian_kde
import plotly.graph_objects as go
import plotly.figure_factory as ff
import warnings
warnings.filterwarnings('ignore')

print("1. Initializing Deep Space Hydrodynamic Radar...")

# -----------------------------------------------------------------------------
# A. LOAD & PREP LOCALIZED DATA
# -----------------------------------------------------------------------------
col_names = [
    "FRN", "Exclusion_Code", "D", "Galaxy_Name", "Distance_Modulus", "Error",
    "Distance_Mpc", "Method", "REFCODE", "Target_ID", "Redshift_z",
    "Hubble_const", "Blank", "Adopted_LMC", "Notes"
]

df_ned = pd.read_csv("/NED30.5.1-D-17.1.2-20200415.csv", skiprows=13, names=col_names, low_memory=False)

for col in ['Redshift_z', 'Distance_Mpc', 'Error']:
    df_ned[col] = pd.to_numeric(df_ned[col], errors='coerce')

df_ned = df_ned.dropna(subset=['Redshift_z', 'Distance_Mpc', 'Galaxy_Name', 'Error'])

# Pristine SNIa Filter (< 5% error) & Expanded 1000 Mpc Boundary
df_sc = df_ned[df_ned['Method'].str.contains('SNIa', na=False, case=False)].copy()
```

```
df_sc['Error_Margin'] = (df_sc['Error'] / df_sc['Distance_Mpc']) * 100

df_local = df_sc[(df_sc['Error_Margin'] <= 5.0) & (df_sc['Distance_Mpc'] <= 1000.0)]


df_roleup = pd.read_csv("/Roleup.csv", low_memory=False)[['Object Name', 'RA', 'Dec']].dropna()

df_unique = pd.merge(df_local, df_roleup, left_on='Galaxy_Name', right_on='Object Name',
how='inner').drop_duplicates(subset=['Galaxy_Name']).copy()


df_unique['X'] = df_unique['Distance_Mpc'] * np.cos(np.radians(df_unique['RA']))

df_unique['Y'] = df_unique['Distance_Mpc'] * np.sin(np.radians(df_unique['RA']))


print(f"   -> Radar locked on {len(df_unique)} deep-space structural nodes.")


# ------------------------------------------------------------------------------

# B. CALCULATE FLUID DELTAS

# ------------------------------------------------------------------------------

print("2. Calculating True Fluid Eddies & Pressure Gradients...")

H0 = 68.1

c = 299792.458

df_unique['Expected_Z'] = (df_unique['Distance_Mpc'] * H0) / c

df_unique['Fluid_Delta'] = df_unique['Redshift_z'] - df_unique['Expected_Z']


# ------------------------------------------------------------------------------

# C. INTERPOLATE WEATHER, NORMALIZE WINDS, & CALCULATE MATTER DENSITY

# ------------------------------------------------------------------------------

print("3. Interpolating Macro-Currents & Physical Sediment Pooling (KDE)...")


grid_x, grid_y = np.mgrid[-1000:1000:400j, -1000:1000:400j]


# 1. Fluid Pressure Grid

grid_z = griddata(

   (df_unique['X'], df_unique['Y']),

   df_unique['Fluid_Delta'],

   (grid_x, grid_y),

   method='cubic'
```

```
)

# 2. Vector Gradient (Cosmic Wind)
dy, dx = np.gradient(grid_z.T)
u_wind, v_wind = -dx, -dy

skip = 8
x_q = grid_x[::skip, ::skip].flatten()
y_q = grid_y[::skip, ::skip].flatten()
u_q = u_wind[::skip, ::skip].flatten()
v_q = v_wind[::skip, ::skip].flatten()

valid = ~np.isnan(u_q) & ~np.isnan(v_q)
x_valid, y_valid, u_valid, v_valid = x_q[valid], y_q[valid], u_q[valid], v_q[valid]

arrow_length = 30.0
magnitude = np.sqrt(u_valid**2 + v_valid**2)
magnitude[magnitude == 0] = 1.0
u_norm = (u_valid / magnitude) * arrow_length
v_norm = (v_valid / magnitude) * arrow_length

# 3. KERNEL DENSITY ESTIMATION (The Matter Overlap Test)
# Calculate the literal physical density of the rendered nodes
positions = np.vstack([df_unique['X'], df_unique['Y']])
kde = gaussian_kde(positions)

# Evaluate the physical density across the grid
grid_coords = np.vstack([grid_x.flatten(), grid_y.flatten()])
density_z = kde(grid_coords).reshape(grid_x.shape)

# Identify the top 10% most dense clusters for rendering
density_threshold = np.percentile(density_z, 90)

# ------------------------------------------------------------------------------
```

```
# D. RENDER THE DASHBOARD
# ------------------------------------------------------------------------------
fig = go.Figure()

# Layer 1: The Weather Radar (Fluid Drag)
fig.add_trace(go.Contour(
    x=grid_x[:,0], y=grid_y[0,:], z=grid_z.T,
    colorscale='RdBu_r', zmin=-0.05, zmax=0.05, opacity=0.5,
    colorbar=dict(title="Fluid Eddy (ΔZ)", x=1.05),
    name='Macro-Current'
))

# Layer 2: The Sediment Pools (Physical Density Hotspots)
fig.add_trace(go.Contour(
    x=grid_x[:,0], y=grid_y[0,:], z=density_z.T,
    colorscale=['rgba(0,0,0,0)', 'lime'], # Transparent to bright green
    showscale=False,
    contours=dict(start=density_threshold, end=np.max(density_z), size=(np.max(density_z)-density_threshold)/3),
    line=dict(width=3, color='lime'),
    name='High-Density Sediment Pools (KDE)'
))

# Layer 3: The Quiver Plot (Normalized Arrows)
fig_quiver = ff.create_quiver(
    x_valid, y_valid, u_norm, v_norm, scale=1, arrow_scale=0.3,
    line=dict(color='cyan', width=1.5), name='Cosmic Wind Vectors'
)
fig.add_traces(fig_quiver.data)

# Layer 4: The Galaxies
fig.add_trace(go.Scatter(
    x=df_unique['X'], y=df_unique['Y'], mode='markers',
    marker=dict(size=3, color='white', opacity=0.9, line=dict(width=0.5, color='black')),
    name='Historical Sediment',
```

```
    text=df_unique['Galaxy_Name'],
    hovertemplate="<b>%{text}</b><br>Fluid Delta: %{customdata:.4f}<extra></extra>",
    customdata=df_unique['Fluid_Delta']
))

# Layer 5: Earth
fig.add_trace(go.Scatter(
    x=[0], y=[0], mode='markers',
    marker=dict(size=12, color='white', symbol='cross', line=dict(width=2, color='black')),
    name='Earth (Anchor Zone)'
))

fig.update_layout(
    title='IBT Deep Space Radar: Vector Flow & Sediment Pooling Verification',
    xaxis_title='X Axis (Mpc)', yaxis_title='Y Axis (Mpc)',
    xaxis=dict(range=[-1100, 1100], showgrid=True, gridcolor='#333', zerolinecolor='white'),
    yaxis=dict(range=[-1100, 1100], showgrid=True, gridcolor='#333', zerolinecolor='white', scaleanchor="x", scaleratio=1),
    plot_bgcolor='black', paper_bgcolor='black', font=dict(color='white'),
    margin=dict(l=40, r=40, b=40, t=60)
)

fig.show()
print("Density check complete. Green topographical contours highlight maximum matter clustering.")
```